

Homework 4

MEGN544/ROBO554 Robot Mechanics: Kinematics, Dynamics, and Control

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Problem 1. Derive from the inverse twist problem equations (p. 34 of notes) the simplified forward and inverse equations for the twist parameterization in the special case of parallel displacement $\mathbf{d}$ with the rotation axis $\hat{\mathbf{k}}$.

Problem 2. Sketch the zero-angle configuration from the following DH table.

Link	a	d	α	θ
1	1	1	$\pi/2$	$\star$
2	0.5	1	$-\pi/2$	$\star$
3	0	0	0	$\star$
4	1	0	$\pi/2$	$\star$
5	0	0	$-\pi/2$	$\star$
6	0	0	$\pi/2$	$\star$

Problem 3. Sketch the zero-angle configuration from the following DH table.

Link	a	d	α	θ
1	1	$\star$	$\pi/2$	0
2	1	$\star$	$\pi/2$	$\pi/2$
3	1	$\star$	0	0

Problem 4. For the manipulator of Problem 2:

- (a) What is the homogeneous transformation ${}^0\mathbf{T}_6$ if all joint angles are 0?
- (b) What is the homogeneous transformation ${}^0\mathbf{T}_6$ if all joint angles are $\pi/6$?

Problem 5. For the manipulator of Problem 3:

- (a) What is ${}^0\mathbf{T}_3$ when $d_1 = 1$ and all other joints are 0?

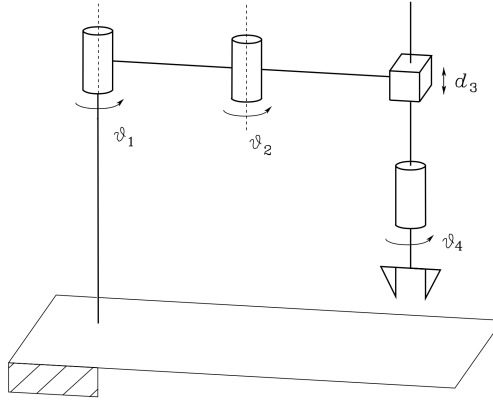


Figure 1: RRPR manipulator.

(b) What is ${}^0\mathbf{T}_3$ when $d_2 = 1$ and all other joints are 0?

(c) What is ${}^0\mathbf{T}_3$ when $d_3 = 1$ and all other joints are 0?

Problem 6. Derive the DH parameters for the RRPR manipulator shown in Fig. 1.